



Flight Delay Prediction with Random Forest

Predicting flight delays for better decision-making.

Project Objectives



1

Predictive Model

Predict flight delays using machine learning.

2

Feature Focus

Flight details, airline, and aircraft type.

3

Real-Time Predictions

Provide timely insights on potential delays.

4

Improved Decision-Making

Better planning for passengers and airlines.

Data Preparation

Data Source

Kaggle Flight Delays dataset.

Feature Engineering

Extracted time-based features and converted boolean columns.

Data Cleaning

Handled missing values and cleaned columns with invalid data.

Preprocessing Methods

1

Numeric Data

Imputed missing values using the median and scaled features.

2

Categorical Data

Imputed missing values using constant fill and applied OneHotEncoding.

3

Final Feature Set

Numeric and categorical features for model training.

Model Selection

RandomForestRegressor

Chosen for its ability to handle complex data and robustness.

Hyperparameters

Number of trees and random state for reproducibility.

Training & Testing

1

Data Split

Training dataset (80%) and testing dataset (20%).

2

Model Training

Trained the Random Forest model on the training data.

3

Evaluation Metrics

MSE and R^2 Score to measure model performance.

Results

Metric	Value
MSE	183.75 minutes ²
R ² Score	0.82

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Example Prediction



Flight Details

Delta, LAX to JFK, Thursday at 1 PM.



Predicted Delay

38.25 minutes.



Conclusions

1

Successful Model

Reliable delay prediction model using flight details.

2

Customer Satisfaction

Improve customer experience with delay predictions.

3

Airline Optimization

Optimize flight schedules and resources.

A black and white photograph of a man with dark hair and glasses, wearing a white lab coat over a dark shirt. He is sitting at a desk, looking at a laptop screen, with his hands on the keyboard. The background is dark and out of focus.

Next Steps

1

Feature Expansion

Include weather conditions and air traffic control data.

2

Model Optimization

Tune hyperparameters for better performance.

3

Web Deployment

Integrate the model into a web application for real-time use.

4

Explore Other Models

Evaluate boosting techniques for enhanced accuracy.